

NSF FMRG Phase Field

José Mancias
Arroyave Group

Phase Field Parameters Source

- $\Gamma = (\sigma T_m) / L_f$
- $\mu = 1/(\beta_0 L_f)$

Table 2

Phase field model parameters for pseudobinary Fe-Cr approximation of 316L steel as an austenite-liquid system.

Equilibrium partition coefficient k_e	0.791*
Equilibrium liquidus slope m_l^e	-3.49 K/wt%*
Alloy concentration c_l^0	17 wt%
Gibbs-Thomson coefficient Γ	3.47×10^{-7} K m
Latent heat L	1.25×10^9 J/m ³ *
Heat capacity C_p	5.46×10^6 J/K/m ³ *
Solutal capillary length d_0	43 nm
Liquid diffusion coefficient D_L	3.0×10^{-9} m ² /s**
Solid diffusion coefficient D_s	0 m ² /s
Kinetic coefficient β_0	0.269 s/m
Capillary anisotropy strength ϵ_c	0.018 [31]
Kinetic anisotropy strength ϵ_k	0.13 [31]
Solute trapping velocity V_D^{PF}	1.0 m/s [30]
Computational interface width W_0	25, 50, or 100 nm
Trapping parameter A	0.528, 0.264, or 0.132
Smallest mesh spacing dx	$0.6W_0$

*: Estimated using Thermo-Calc's TCFE9 database [25].

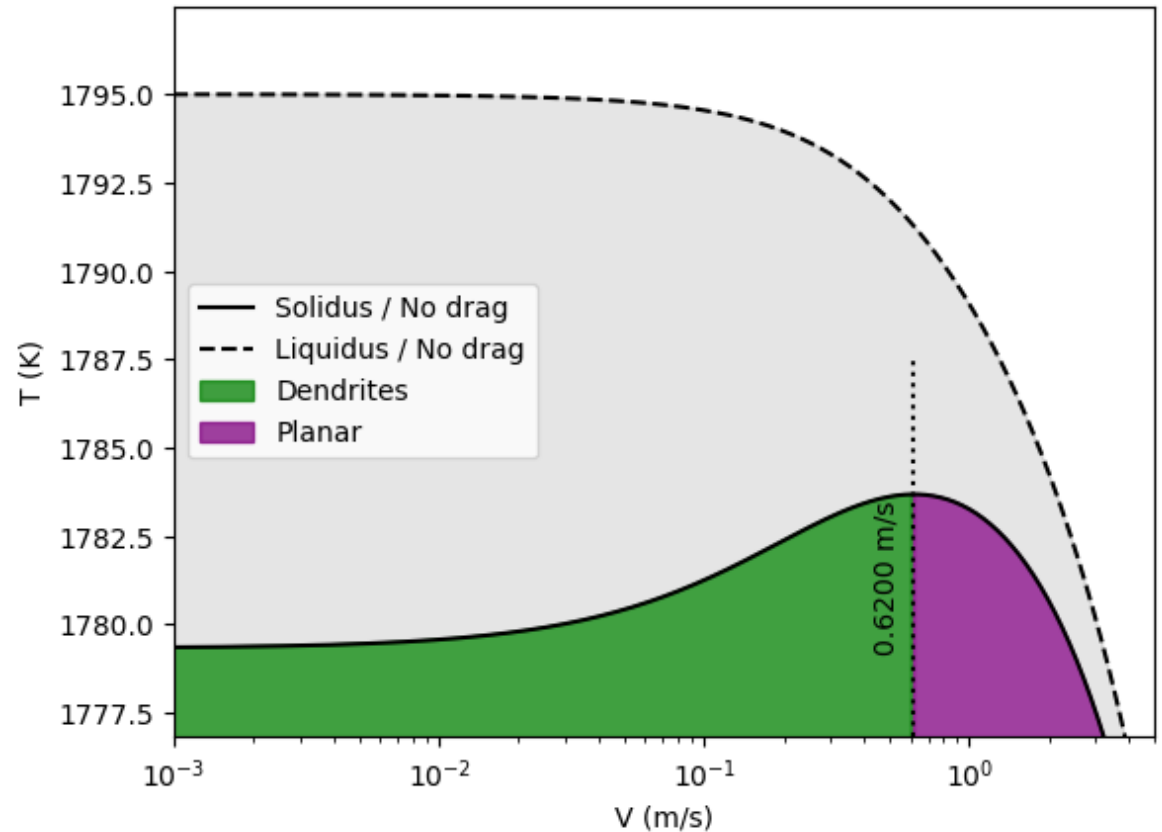
**: Estimated using Thermo-Calc's MOBFE4 database [25].

Solidification Rate (R) Expectations

- For dilute alloys ($\approx$ linear phase diagrams):

Solidus
$$T_S(V) = T_M + \sum_i \left(\frac{m^{(i)} c_0^{(i)}}{k^{(i)}} \right) - \frac{V}{\mu}$$

Liquidus
$$T_L(V) = T_M + \sum_i \left(m^{(i)} c_0^{(i)} \right) - \frac{V}{\mu}$$

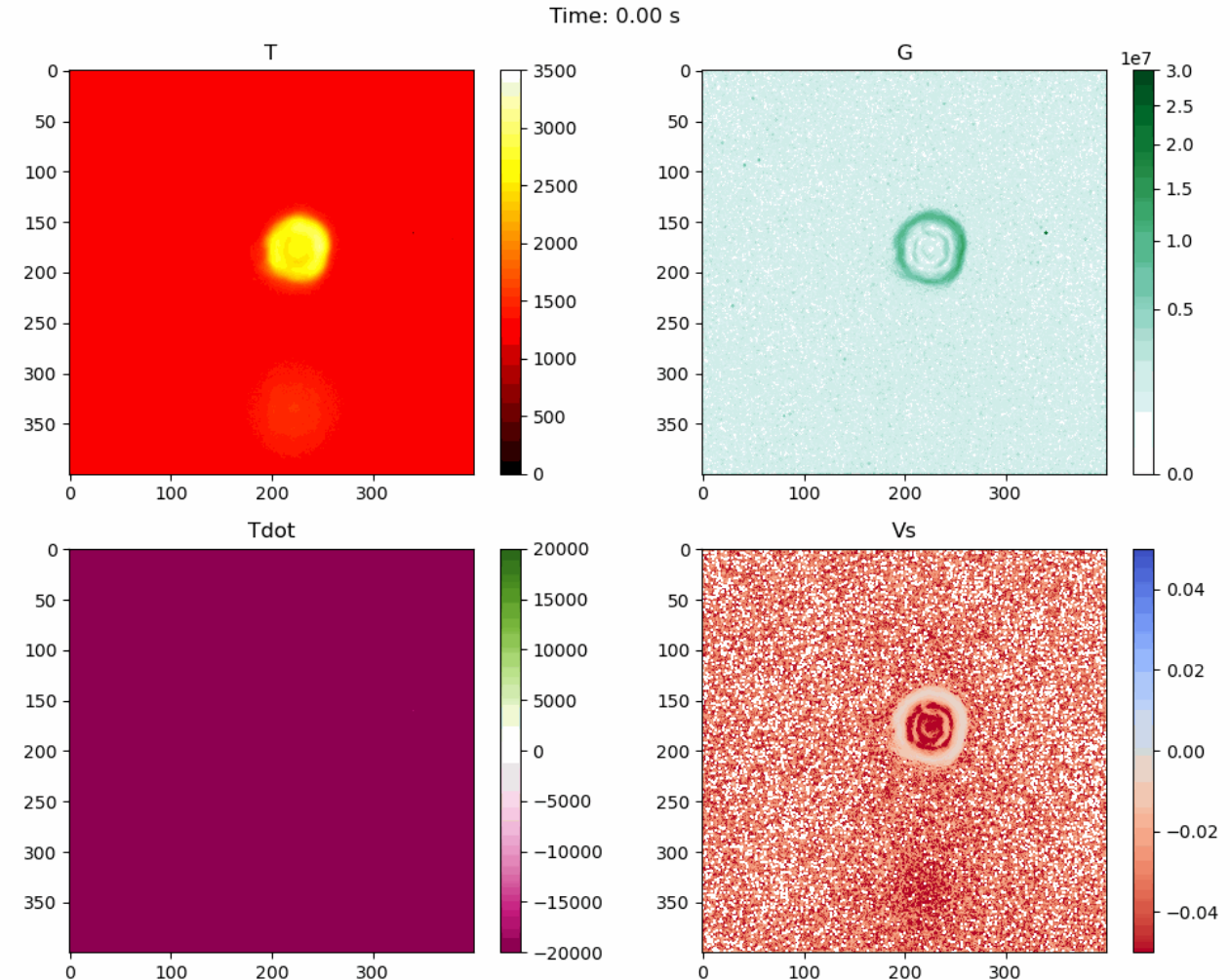


Experimental Temperature Visualization of 316L

$$G = ||\vec{\nabla}T||$$

- Visualize the meltpool
- Calculate Tdot via finite difference
- Tdot defined as negative while heating and positive while cooling

$$\dot{T} = \frac{dT}{dt}$$

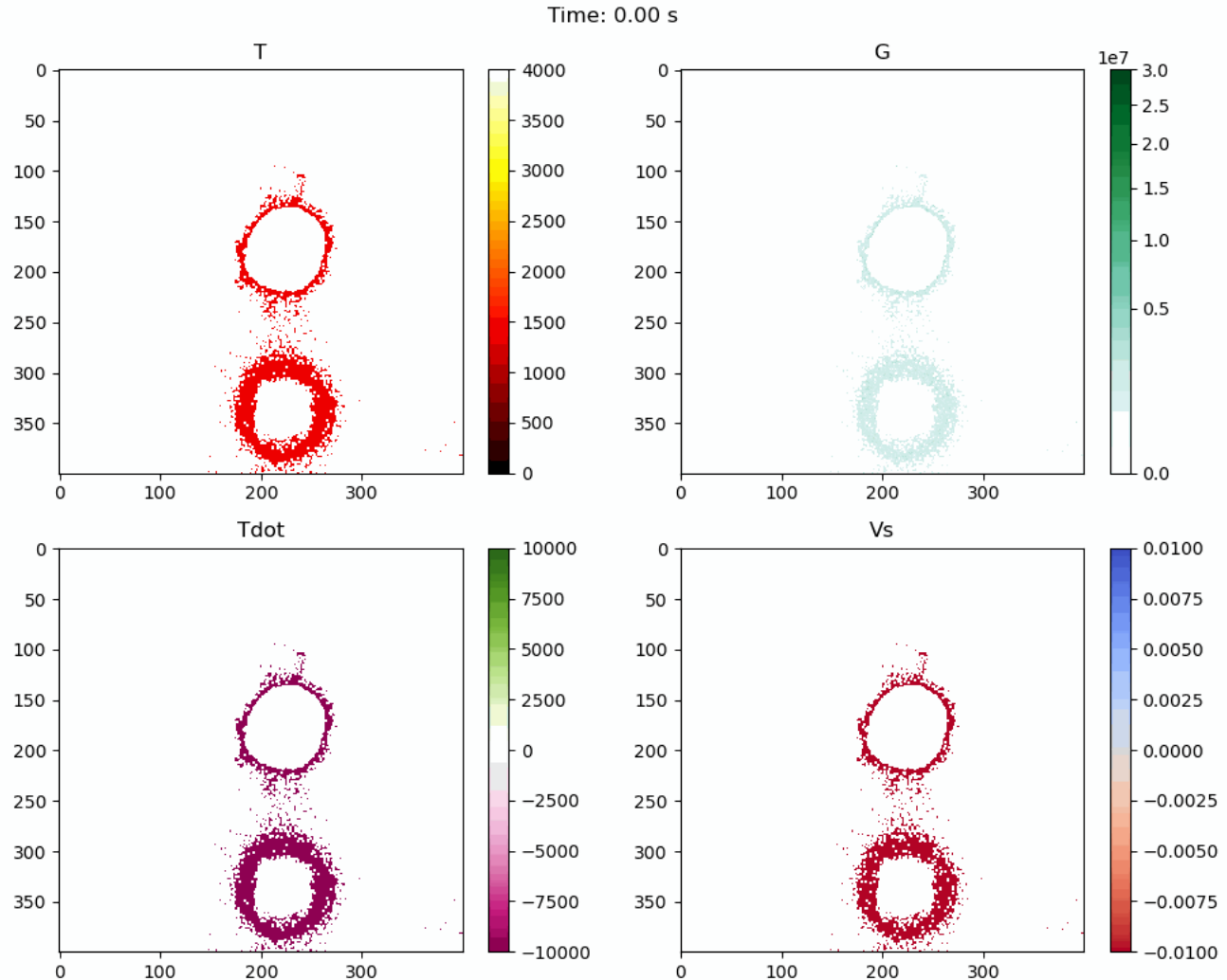


Experimental Temperature Visualization of 316L

- $T = T_s$ isotherm
- Temperature conditions as material solidifies

$$G = ||\vec{\nabla}T||$$

$$\dot{T} = \frac{dT}{dt}$$



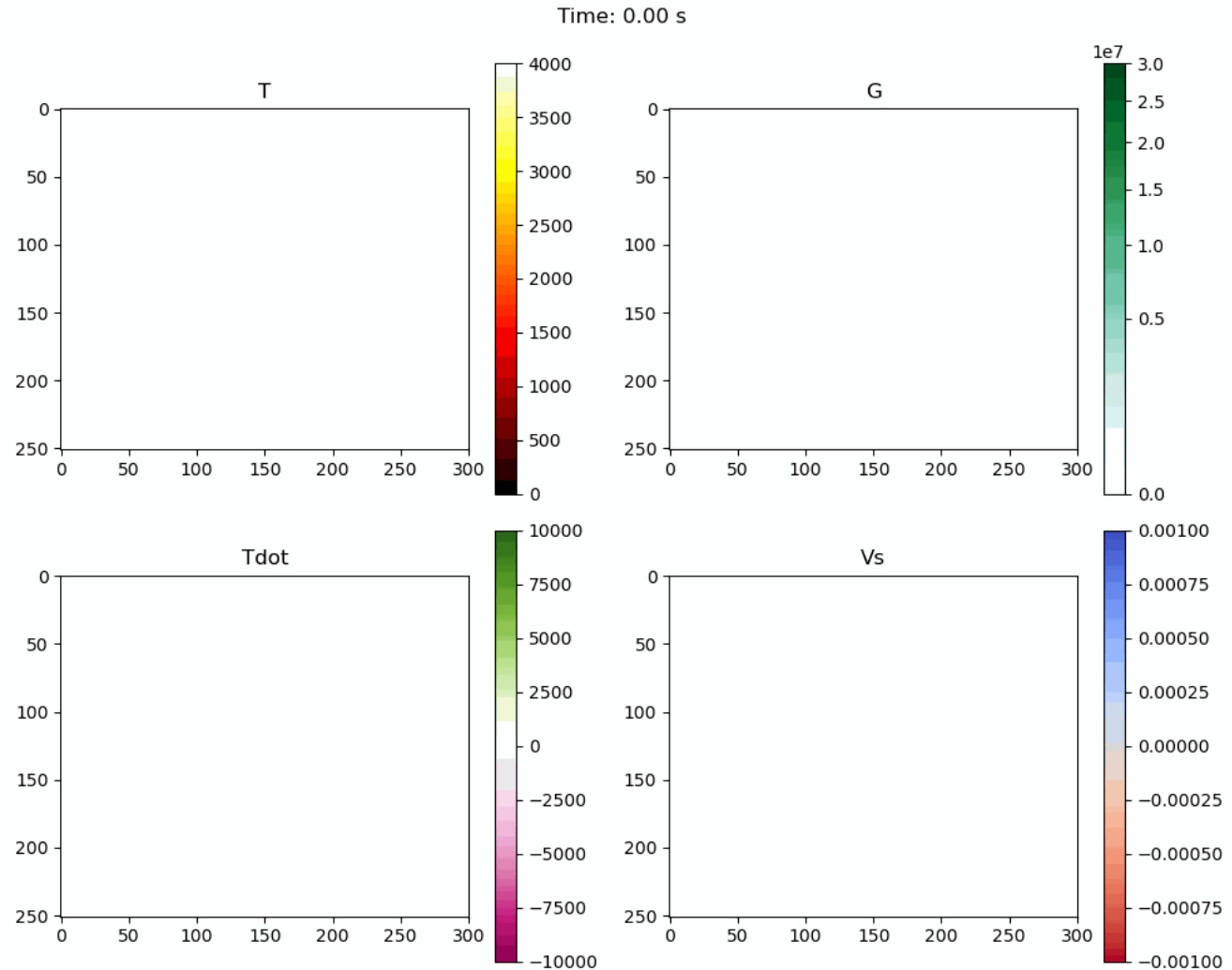
$$V_s = \frac{\dot{T}}{G}$$

Simulated Temperature Visualization of 316L

$$G = ||\vec{\nabla}T||$$

- Simulated temperature using ellipsoid calculation
- More stable results allow us to compare to experimental and see general range of G, Vs
- T = Ts isotherm

$$\dot{T} = \frac{dT}{dt}$$



$$V_s = \frac{\dot{T}}{G}$$

Phase Field Model (Karma 2023)

- Phase Field is a diffuse interface model based on Landau Theory
- Incorporates non-equilibrium effects
 - Velocity dependent $k(V)$ and $m(V)$
 - Reflects solute trapping and interface thickness at high velocities
- Interface thickness enhancement $S = W/W_0$
 - Solute trapping caused by larger S values is mitigated using enhanced solute diffusivity, $q(\phi)$
 - Thermodynamic consistency by including sharp-interface limits in Phase Field
- Advantages
 - Computationally efficient with large values of S
 - Quantitative predictions
 - Predicts solute trapping behavior
 - Captures microstructural transitions

Phase Field Kinetic Equation

$$\tau(n) \frac{\partial \phi}{\partial t} = \nabla \cdot [W(n)^2 \nabla \phi] + \phi - \phi^3 + \sum_{i=x,y} \left[\partial_i \left(|\nabla \phi|^2 W(n) \frac{\partial W(n)}{\partial (\partial_i \phi)} \right) \right] - \lambda g'(\phi) \left[c + \frac{T - T_M}{m_e} e^{b(1+g(\phi))} \right]$$

Concentration Kinetic Equation

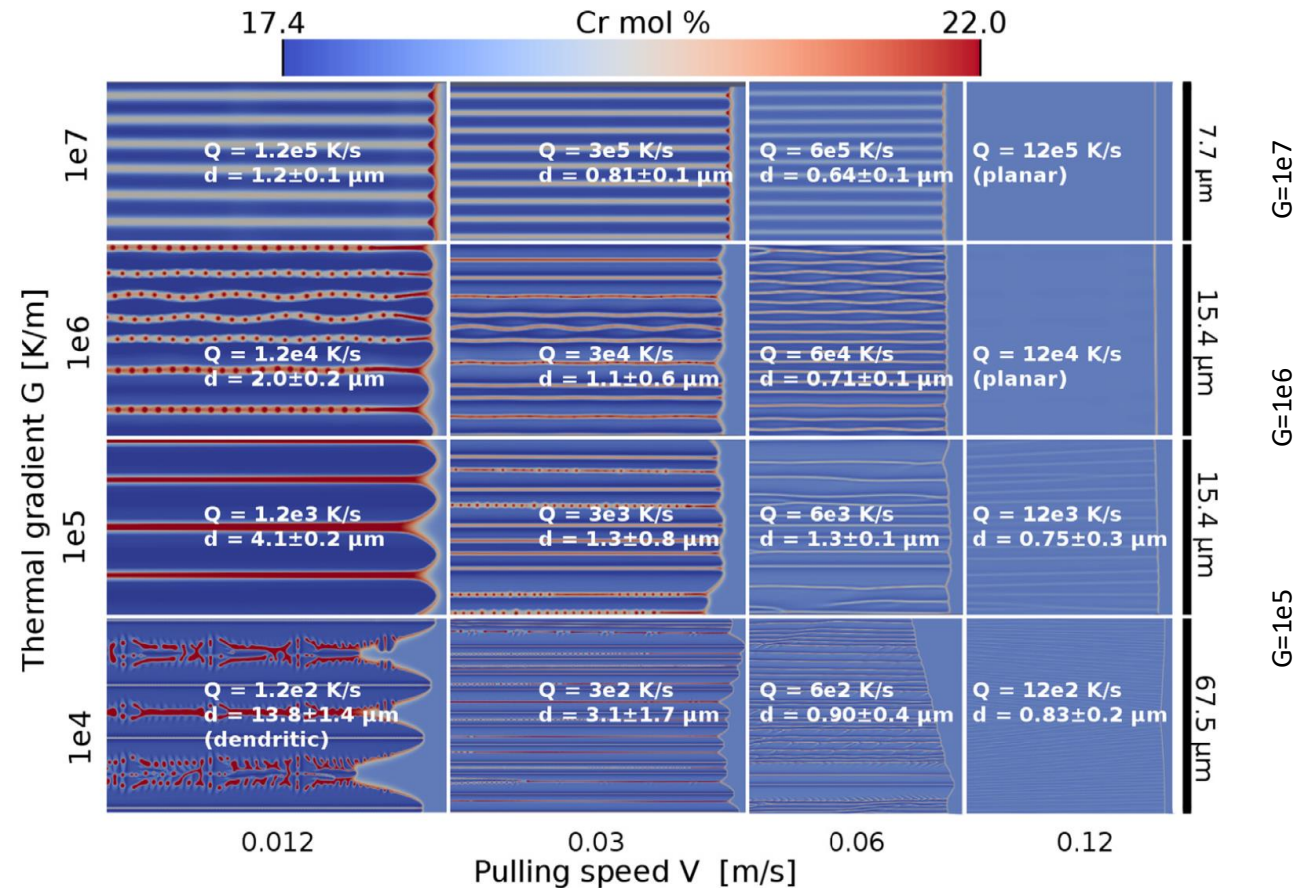
$$\frac{\partial c}{\partial t} = \nabla \cdot \{ D_l q(\phi) c \nabla [\ln c - b g(\phi)] \}$$

Definition of Terms

ϕ - phase field variable	W – interface thickness
c – composition	g – interpolation function
τ - interface time constant	D – diffusivity in liquid

G-R Space FeCr

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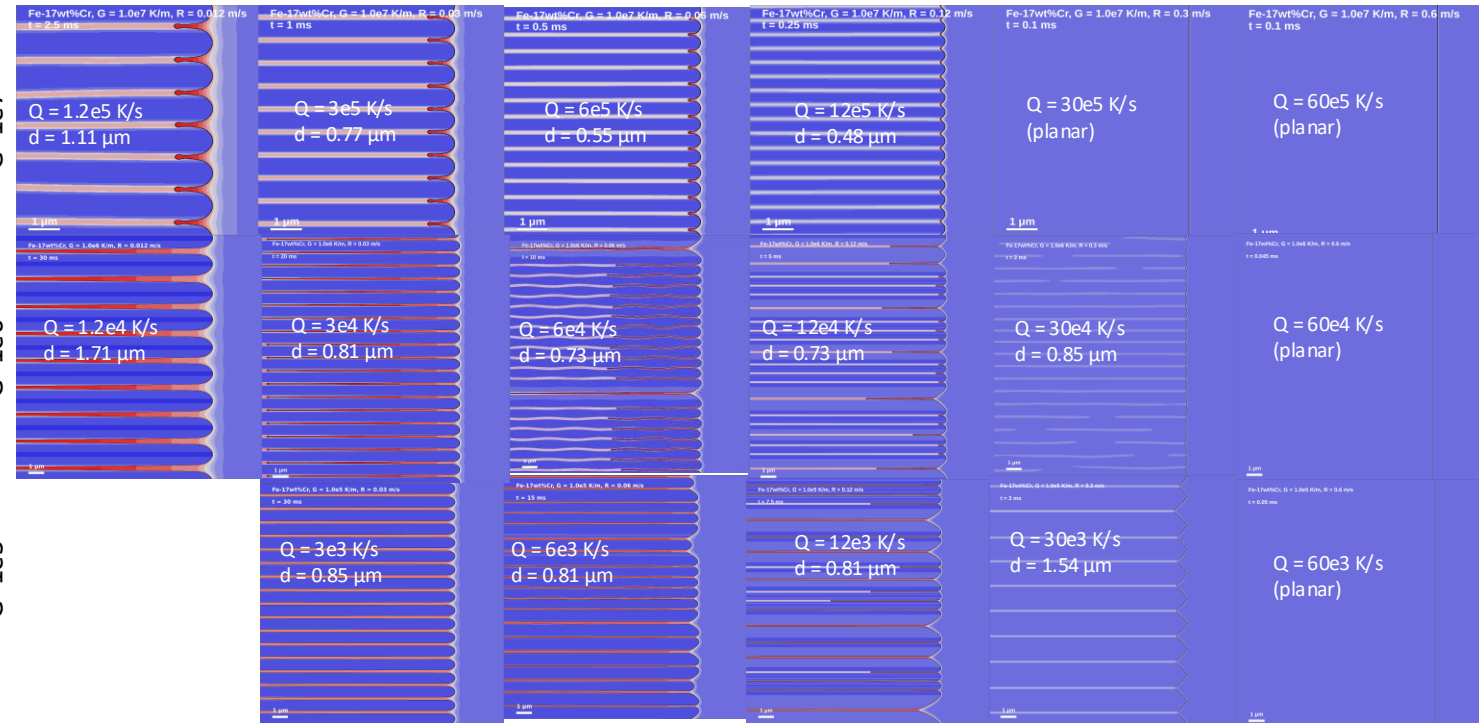
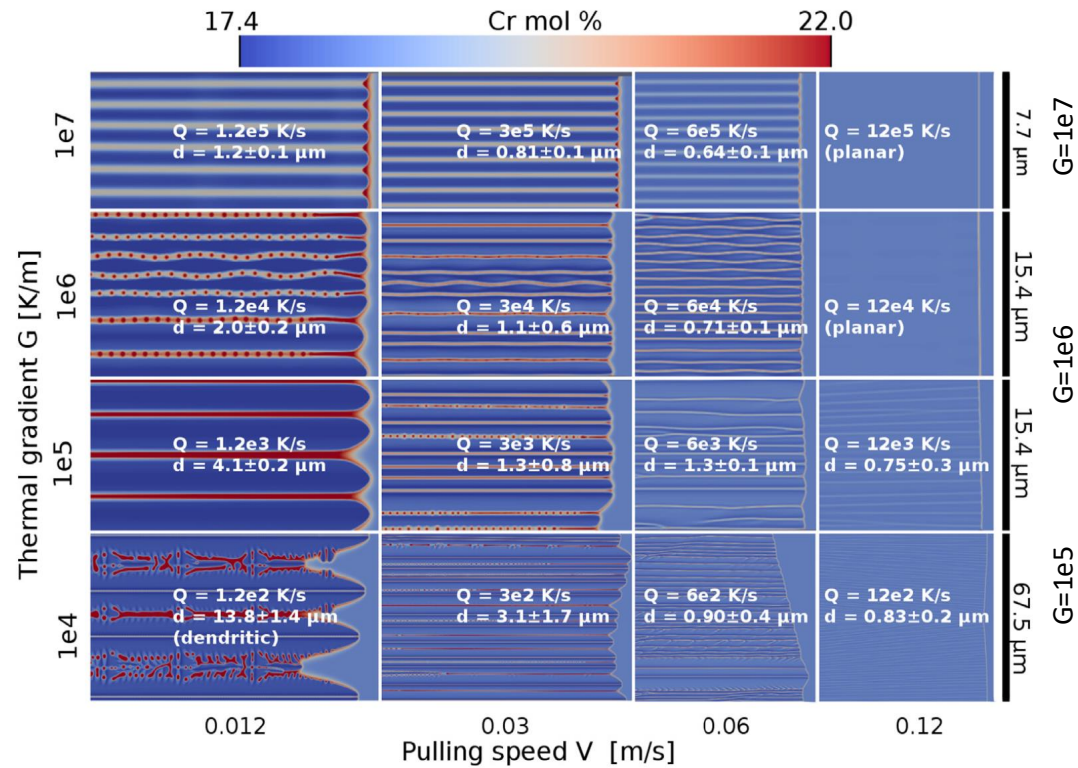
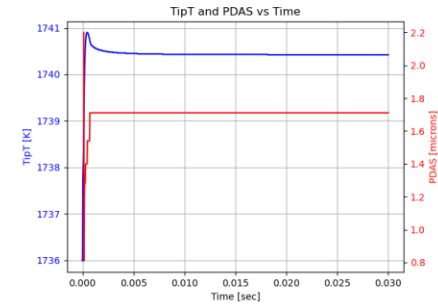


- What can PF provide to a ML model?
 - PDAS
 - Local concentration
- Survey of G and R conditions
- Assumptions:
 - 1D thermal gradient
 - 316L – binary approx - FeCr
- Unfit PF model for rapid solidification

G-R Space FeCr – Karma Model Results

Pinomaa et al 2019

Karma Model Results, $S=5$, $c_0 = 0.17$



$V=0.012$

$V=0.03$

$V=0.06$

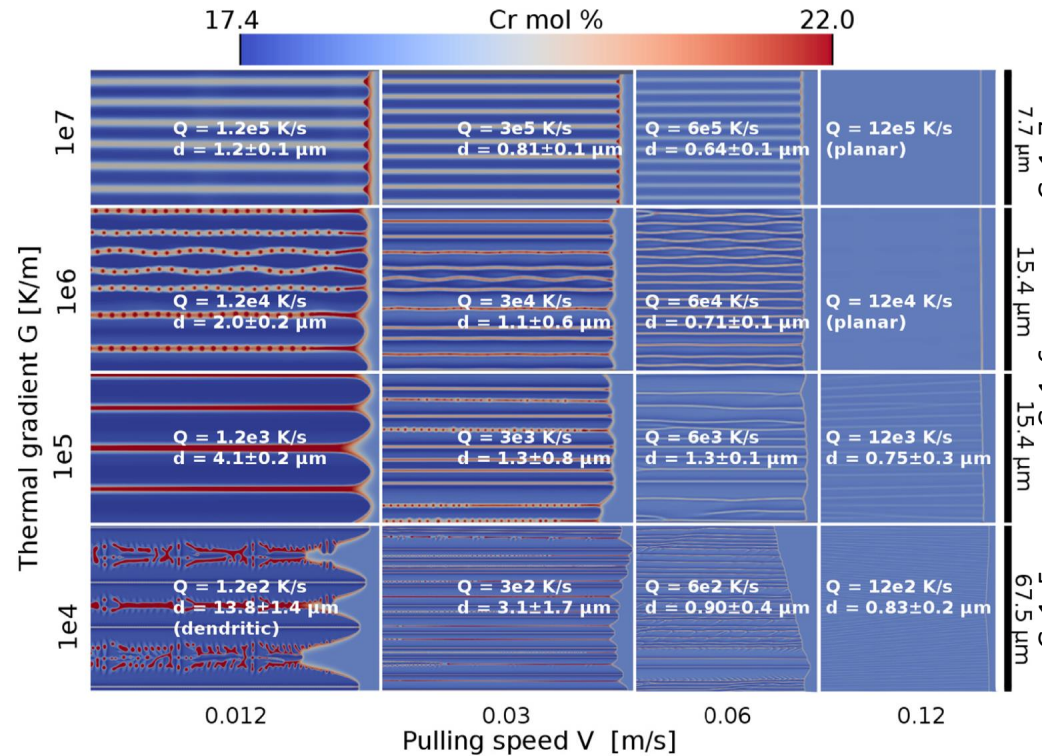
$V=0.12$

$V=0.3$

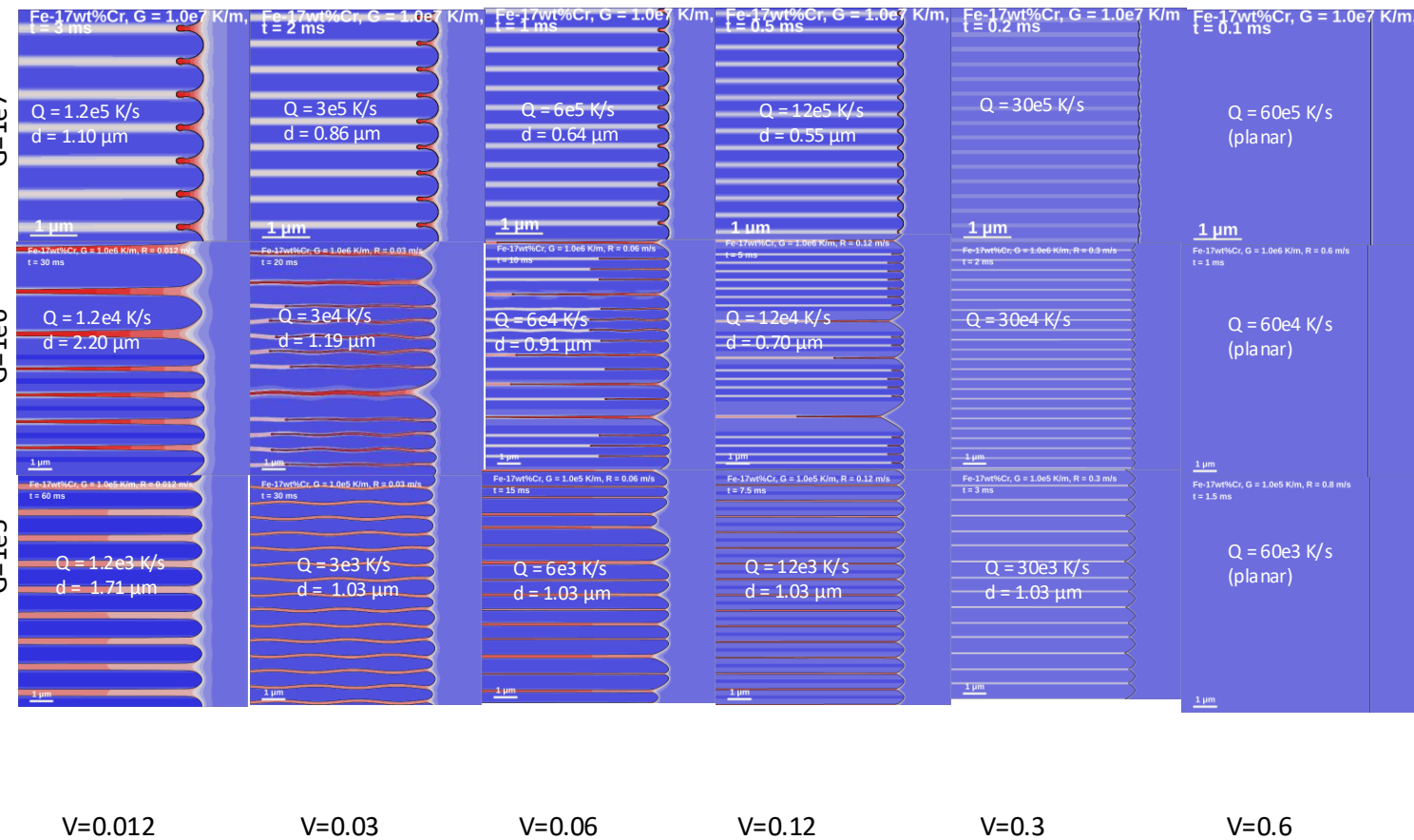
$V=0.6$

G-R Space FeCr – Karma Model Results

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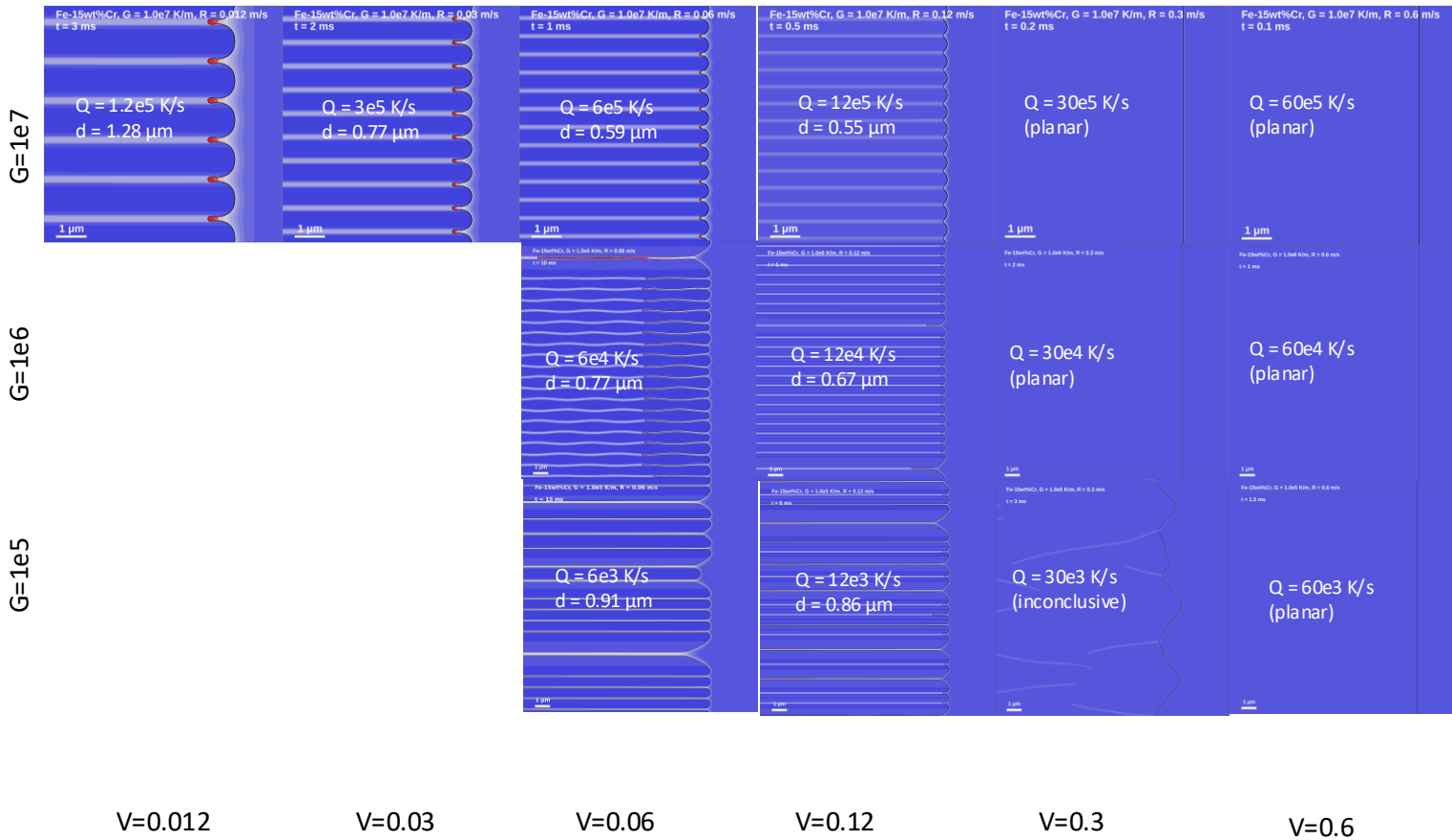


Karma Model Results, $S=8$, $c_0 = 0.17$



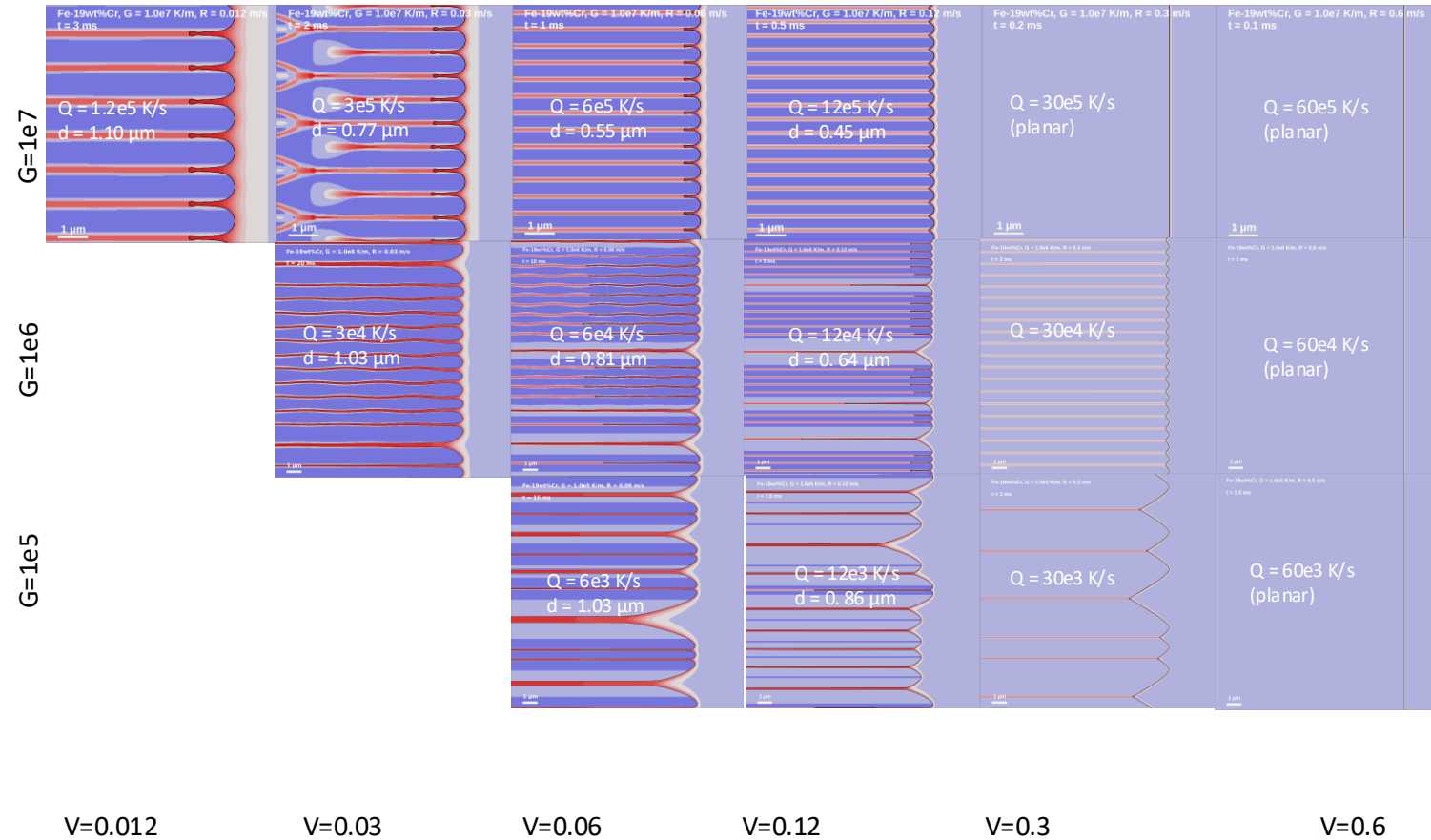
G-R Space FeCr – Karma Model Results

Karma Model Results, $S=5$, $c_0 = 0.15$



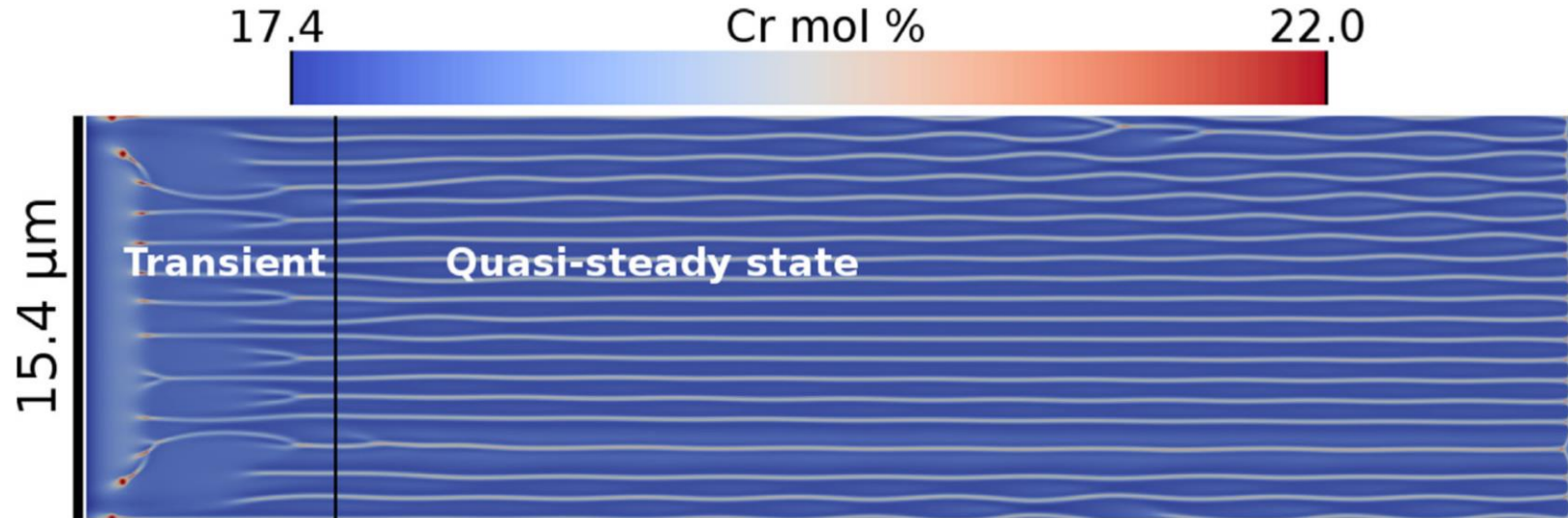
G-R Space FeCr – Karma Model Results

Karma Model Results, $S=5$, $c_0 = 0.19$

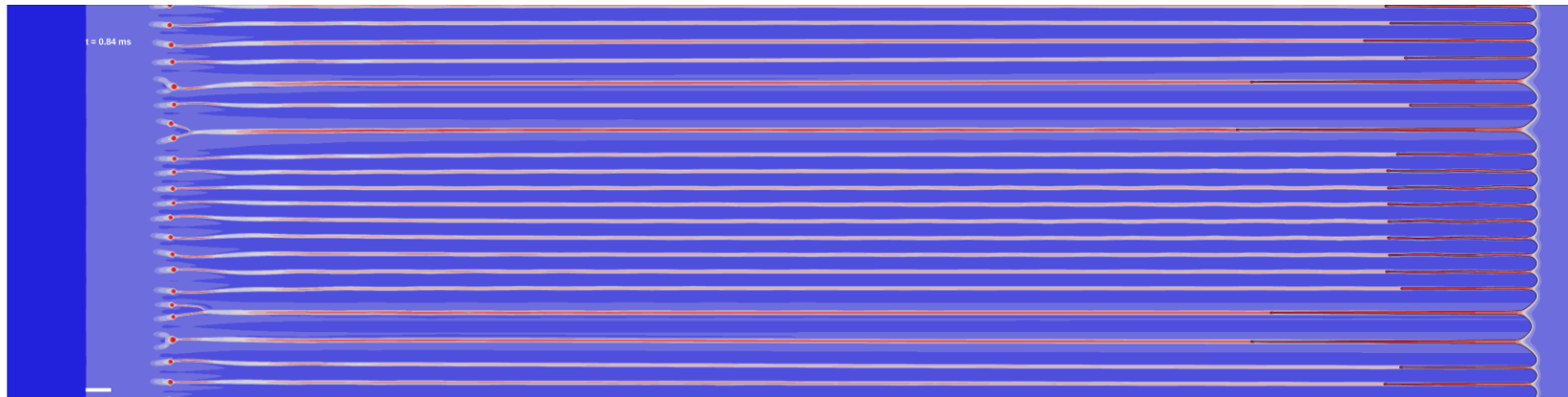


Long form $G=1e6$, $C=0.15$, $S=5$, $R=0.06$

Pinomaa 2019



Karma Model



PDAS vs Q Karma comparison to Pinomaa

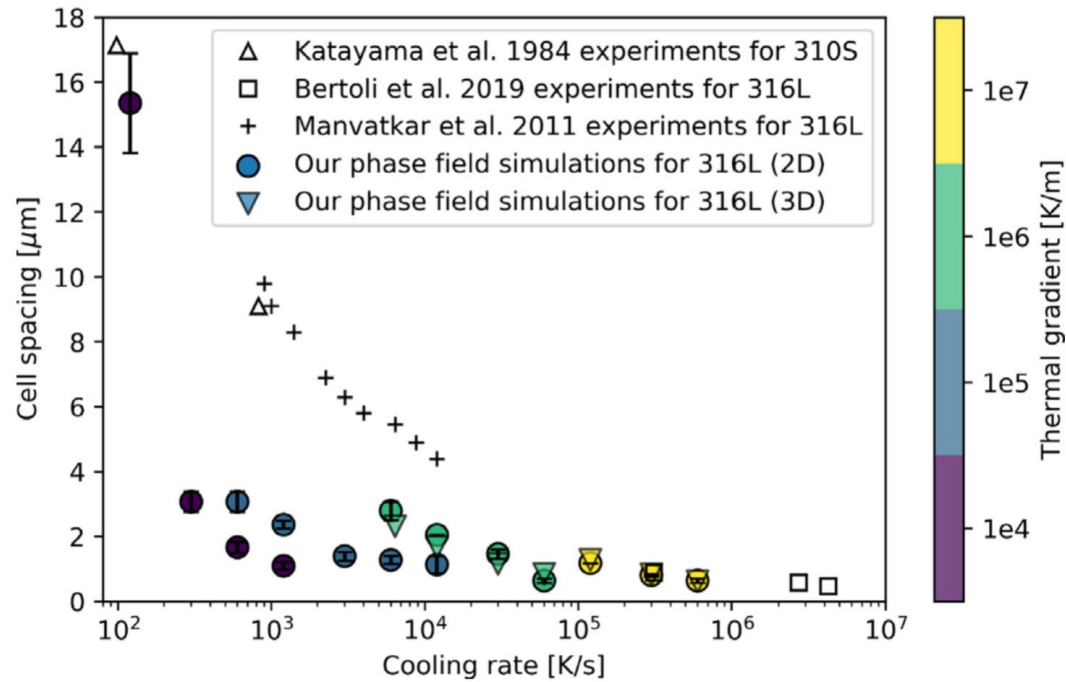
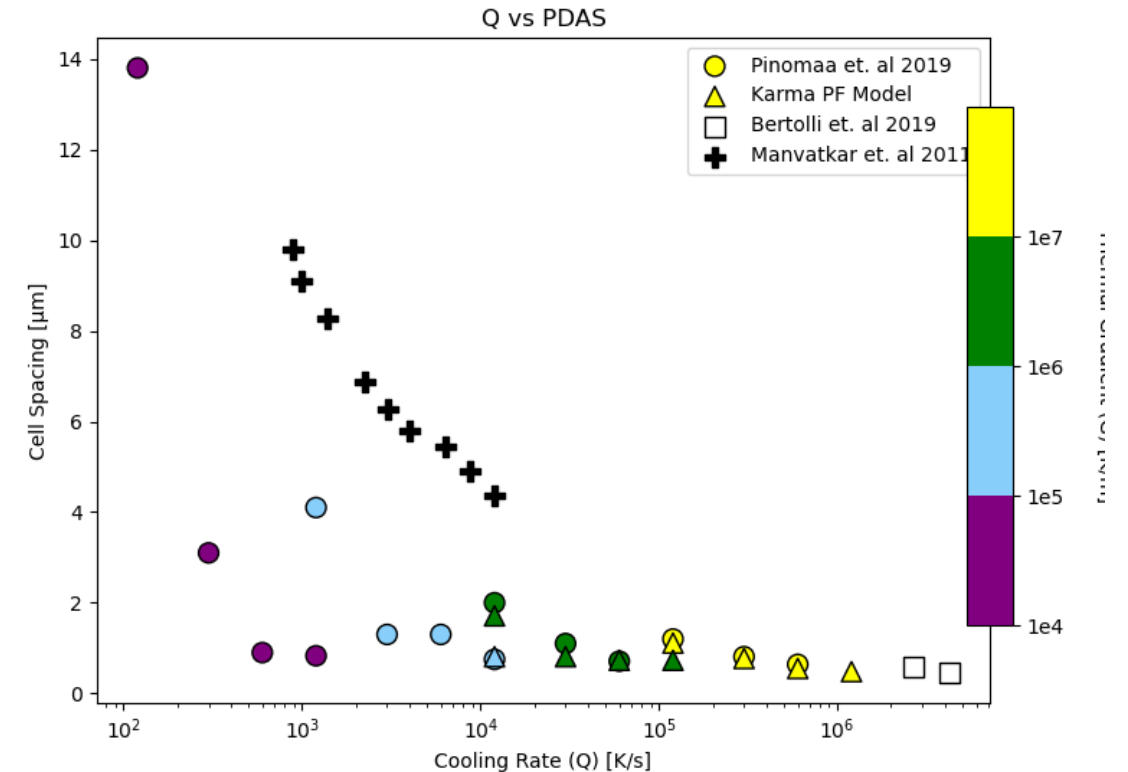
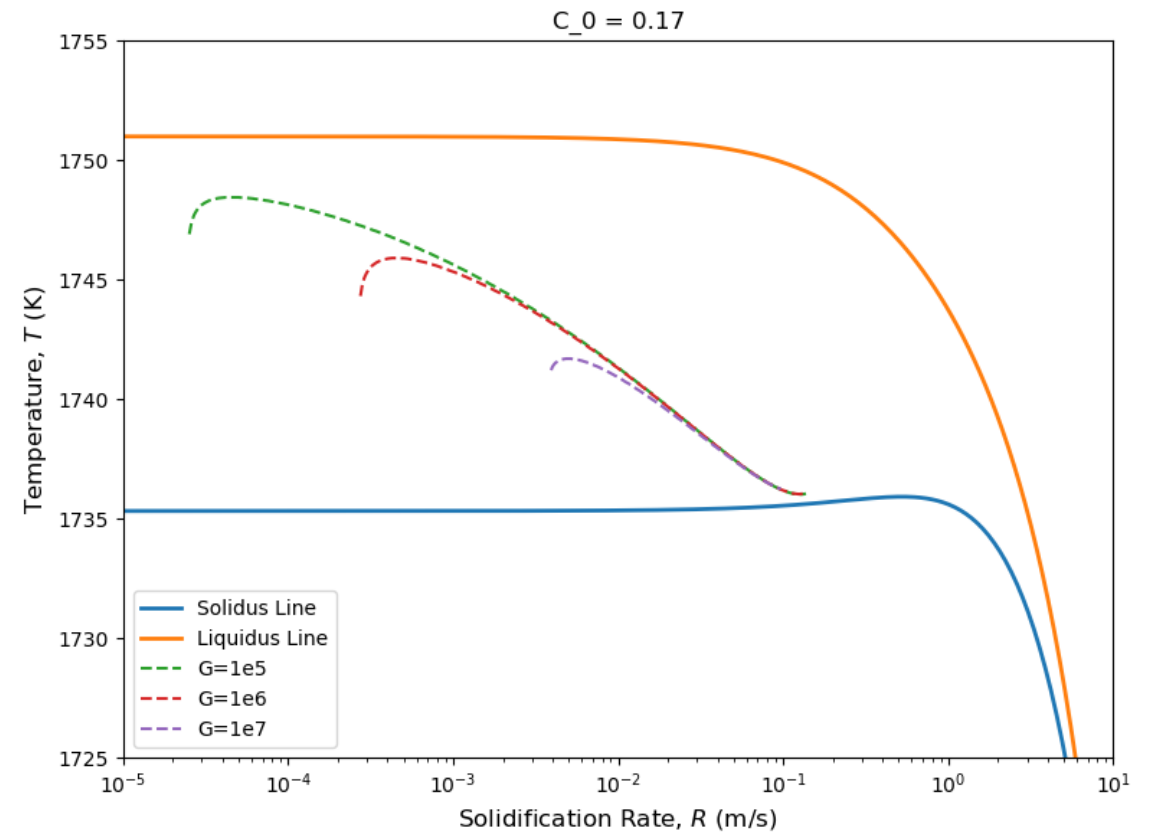
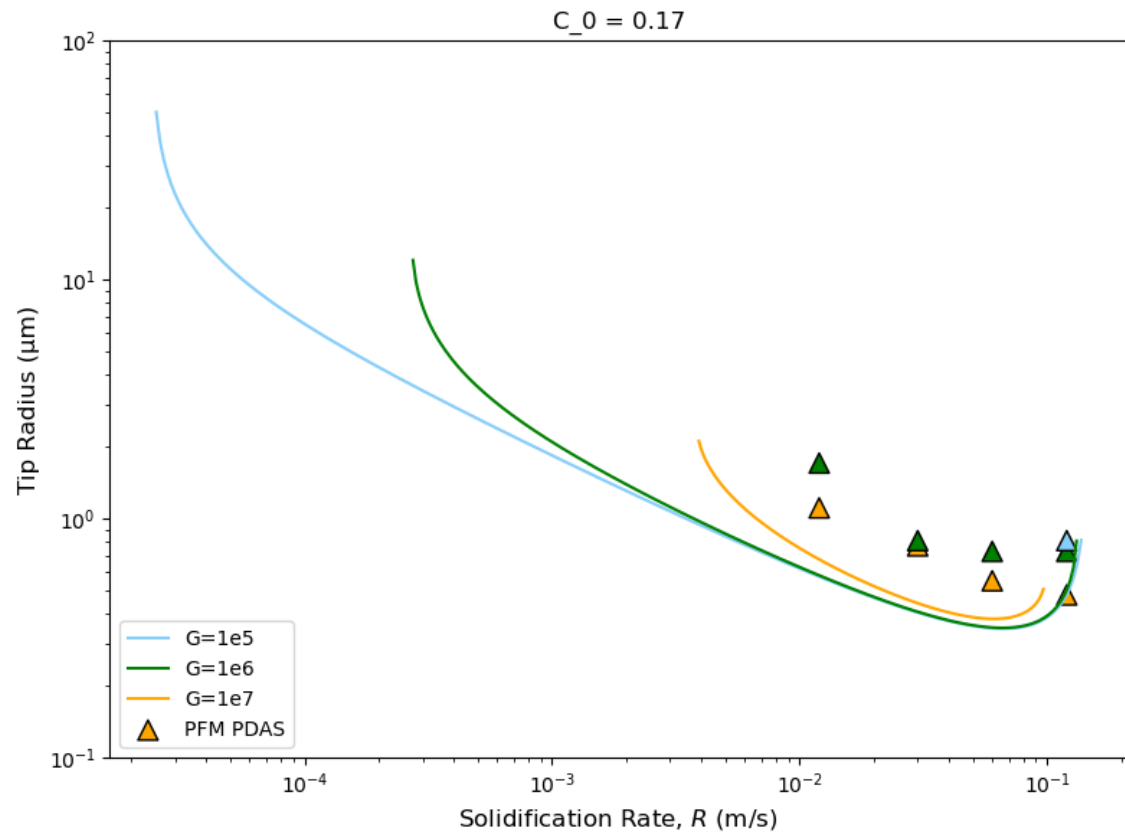


Fig. 8. Average cell spacing versus cooling rate in our simulations (circles in 2D, downward-triangle in 3D) colored by thermal gradient G , and experimental data from literature (hollow upward-triangles [22], hollow squares [15], plus signs [49]).



KGT Plots

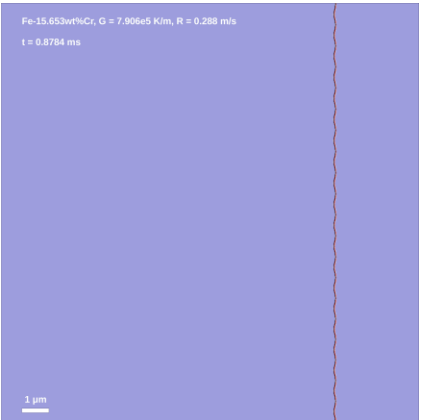
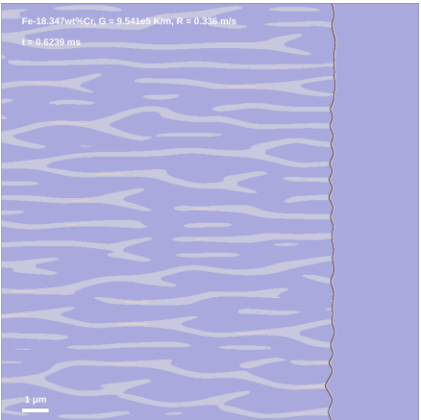
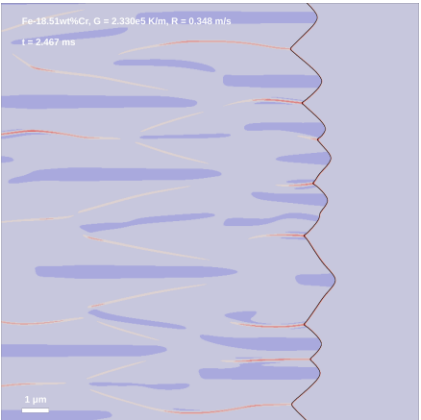


Batch 1



J0	J1	J2	J3	J4	J5	J6	J7	J8	j9
C=16.959	17.122	15.98	18.51	17.449	15.408	18.347	16.469	18.265	15.653
G=3.56e6	4.10e5	7.91e5	8.69e5	1.27e6	2.12e5	3.56e6	1.76e5	1.93e5	4.29e6
R=0.252	0.408	0.312	0.396	0.372	0.372	0.276	0.408	0.432	0.228
Cellular	Planar	Planar	Inconclusive	Planar	Planar	Cellular	Planar	Planar	Cellular

Batch 2



J0	J1	J2	J3	J4	J5	J6	J7	J8	j9
C=15.490	17.286	15.571	18.510	18.347	18.347	17.041	15.653	16.959	16.633
G=3.907e6	1.456e5	1.758e5	2.330e5	3.907e6	9.541e5	1.842e6	7.906e5	4.095e5	5.690e6
R=0.276	0.336	0.300	0.348	0.300	0.336	0.312	0.288	0.324	0.276
Planar			inconclusive	Cellular	inconclusive	Cellular	Planar	inconclusive	Planar

Next Steps

- Visualize 3d space of composition, G, R
 - Planar/Dendritic Transition
- Planar/Dendritic Transition found using batch bayesian